

ASSESSMENT

27 March 2025



Send Your Feedback

Contacts

Tom Collet
Sustainable Finance Analyst
tom.collet@moody's.com

Alice Presotto
Sustainable Finance Associate
alice.presotto@moody's.com

Adriana Cruz Felix
SVP-Sustainable Finance
adriana.cruzfelix@moody's.com

Veolia Environnement S.A.

Second Party Opinion – Green Financing Framework Assigned SQS2 Sustainability Quality Score

Summary

We have assigned an SQS2 Sustainability Quality Score (very good) to Veolia Environnement S.A.'s (Veolia) green financing framework, dated March 2025. Veolia has established a use-of-proceeds framework with the aim of financing projects across three eligible green categories: energy, water and waste. The framework is aligned with the four core components of the International Capital Market Association's (ICMA) Green Bond Principles (GBP) 2021 (including the June 2022 Appendix 1), and the Asia Pacific Loan Market Association, the Loan Market Association, and the Loan Syndications and Trading Association's (LMA/APLMA/LSTA) Green Loan Principles (GLP) 2023. The framework demonstrates a significant contribution to sustainability.

Sustainability quality score



Alignment with principles USE OF PROCEEDS

Overall alignment



FACTORS

ALIGNMENT

Use of proceeds	
Evaluation and selection	
Management of proceeds	
Reporting	

Contribution to sustainability

Final contribution to sustainability



Preliminary contribution to sustainability

Relevance and magnitude

Additional considerations No adjustment

POINT-IN-TIME ASSESSMENT

Scope

We have provided a Second Party Opinion (SPO) on the sustainability credentials of Veolia's green financing framework, including the framework's alignment with the ICMA's GBP 2021 (including the June 2022 Appendix 1) and the LMA/APLMA/LSTA's GLP 2023. Under its framework, Veolia plans to issue use-of-proceeds instruments with the aim of financing projects in three categories: energy, water and waste, as outlined in Appendix 3 of this report.

Our assessment is based on the last updated version of the framework received on 27 March 2025, and our opinion reflects our point-in-time assessment¹ of the details contained in this version of the framework, as well as other public and nonpublic information provided by the company.

We produced this SPO based on our [Assessment Framework: Second Party Opinions on Sustainable Debt](#), published in March 2025.

Issuer profile

Veolia Environnement S.A. (Veolia), headquartered in France, is one of the largest global providers of environmental services, operating in the water, waste and energy sectors. The company supplies drinking water to 113 million people and wastewater treatment services to 103 million people. It also provides waste management and energy services worldwide. In 2023, Veolia employed nearly 218,000 people and reported revenue of €45.4 billion. Veolia is a publicly traded company with a diverse ownership structure, including institutional and individual shareholders. Veolia's 2024-2027 Strategic Program focuses on affordable solutions for depollution, decarbonization and resource regeneration, emphasizing sustainable development and employee engagement.

Strengths

- » The company refers to the EU Taxonomy's substantial contribution criteria for all of the projects financed under this framework.
- » The framework is aligned with identified best practices for three pillars of the green bond principles.
- » An important portion of proceeds will be allocated to the water category which is reaching the highest scores.

Challenges

- » The financing of fossil fuel powered trucks for waste collection entails continuous GHG emissions.
- » There is no independent verification of the impact reporting on environmental benefits associated with the financed projects.

This publication does not announce a credit rating action. For any credit ratings referenced in this publication, please see the issuer/deal page on <https://ratings.moody.com> for the most updated credit rating action information and rating history.

Alignment with principles

Veolia's green financing framework is aligned with the four core components of the ICMA's GBP 2021 (including the June 2022 Appendix 1) and the LMA/APLMA/LSTA's GLP 2023. For a summary of alignment with principles scorecard, please see Appendix 1.

- | | |
|--|--|
| ☒ Green Bond Principles (GBP) | ☒ Green Loan Principles (GLP) |
| ☐ Social Bond Principles (SBP) | ☐ Social Loan Principles (SLP) |
| ☐ Sustainability-Linked Bond Principles (SLBP) | ☐ Sustainability Linked Loan Principles (SLLP) |

Use of proceeds



Clarity of the eligible categories – BEST PRACTICES

Veolia has clearly communicated the nature of the expenditures, and has set eligibility and exclusion criteria for all the eligible categories. The benefits have been clearly identified. The company disclosed that the eligible projects can be located in any area where Veolia operates, spanning the following geographic zones: France and Special Waste Europe, Central and Eastern Europe, Northern Europe, Asia-Pacific, Iberia and Latin America, North America, Italy and Africa and Middle East.

Under Veolia's framework, a limited portion of proceeds is allocated toward equity investment in the form of acquisition of shares in companies that derive at least 90% of their revenue from activities that adhere to the eligibility criteria defined in the framework. The remaining 10% of their revenue must not come from activities listed in the exclusion criteria.

We consider this a nonstandard use of proceeds, which creates certain challenges in terms of asset-level adherence to sustainability objectives, allocation and traceability; impact reporting; and increased risk of double counting. However, because the expenditure related to equity will account for less than 10% of the total net proceeds issued under its framework and only in companies in which Veolia obtains a controlling share, we consider the structure aligned with the spirit of the ICMA's principles and current market practices.

Clarity of the environmental objectives – BEST PRACTICES

Veolia has clearly outlined the environmental objectives across categories. They are relevant for the eligible categories and are coherent with recognized international standards, including the EU Taxonomy and the United Nations' (UN) Sustainable Development Goals (SDGs).

Clarity of the expected benefits – BEST PRACTICES

Veolia has identified relevant environmental benefits for all the eligible categories. The expected benefits are measurable and will be quantified in the impact report. The issuer has identified the expected benefits for all categories. Only refinancing is expected for the first issuance and the estimated share of refinancing will be shared prior to other issuances. The maximum look-back period is 24 months.

Process for project evaluation and selection

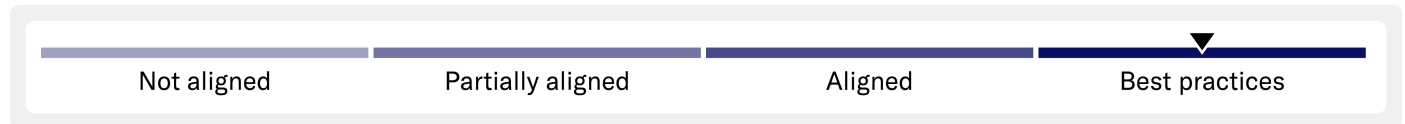


Transparency and clarity of the process for defining and monitoring eligible projects – BEST PRACTICES

Veolia has established a clear and structured decision-making process for determining the eligibility of projects and assets, which is detailed in the publicly available framework. An ESG committee is in place, which comprises functional departments responsible for environmental, social, societal, compliance and governance matters, and ensures their coordination. A process to identify and manage

E&S risks is in place and will be made publicly available as part of the framework. The committee is co-chaired by the group general counsel, the senior executive vice president for strategy and innovation, and the senior executive vice president for human resources. The committee is in charge of reviewing annually the list of eligible projects against the eligibility and exclusion criteria. The tracking of the projects will be done via an internal tool, and minutes of each ESG committee will be recorded. If a project no longer meets the eligibility criteria, it will be substituted. Eligible projects adherence to the eligibility criteria will be monitored until maturity of the green financing instruments. If controversies are identified, they will be discussed and addressed by the ESG Committee.

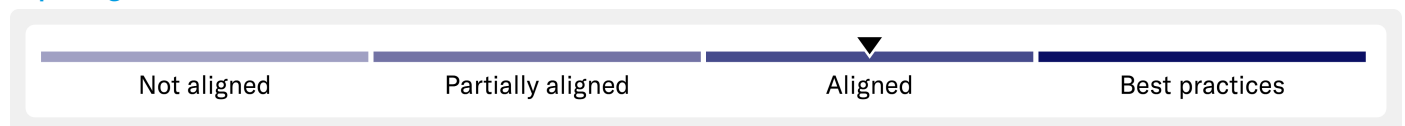
Management of proceeds



Allocation and tracking of proceeds – BEST PRACTICES

Veolia has clearly defined the process for the allocation and tracking of proceeds in its framework, which will be held in the general treasury. The process for proceeds management will be transparently disclosed to the public as part of its framework. The company has committed to fully allocate the proceeds within two years from the date of issuance. Any unallocated proceeds will be invested in cash, cash equivalents or other liquid marketable investments. Veolia will ensure that the portfolio of eligible projects exceeds the issuance amount. The tracked proceeds will be adjusted twice a year by the treasury department.

Reporting

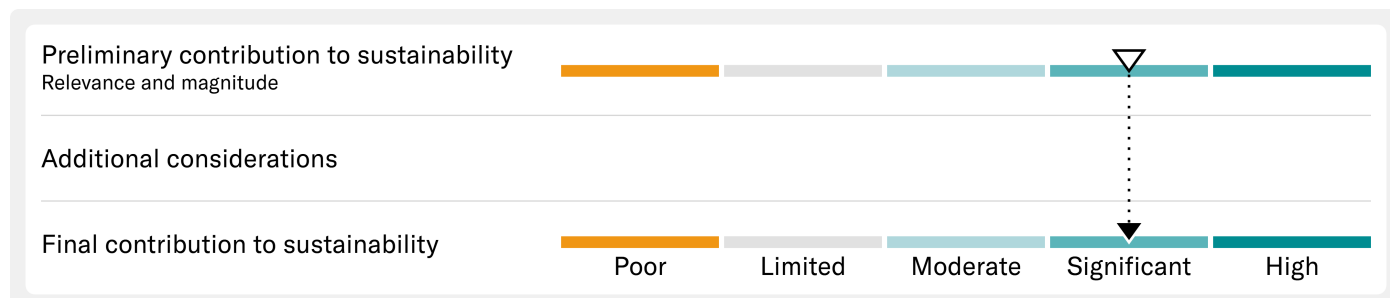


Reporting transparency – ALIGNED

Veolia has committed to provide annual allocation and impact reporting until full allocation, and in the event of material developments while the instruments are outstanding. The report will be publicly available on the company's website and will cover relevant information about the allocation of proceeds and the expected environmental benefits of the projects. The company has identified relevant indicators for each project sub-category. Calculation methodologies and assumptions for the impact indicators will be shared in the impact report. The allocation report will be subject to an external review on an annual basis. However, there will not be an independent assessment of the impact indicators.

Contribution to sustainability

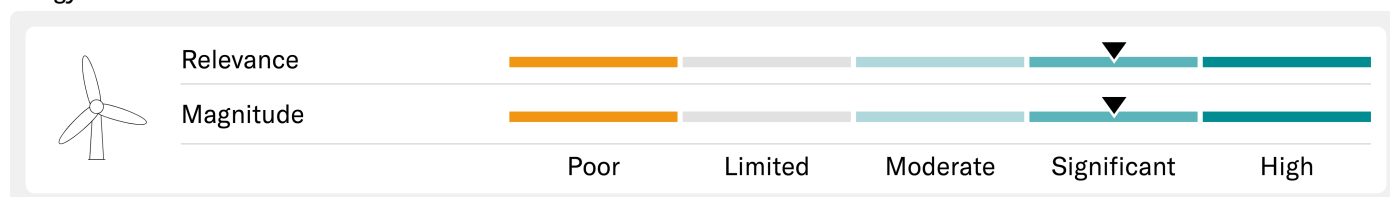
The framework demonstrates a significant overall contribution to sustainability. This reflects a preliminary contribution to sustainability score of significant, based on the relevance and magnitude of the eligible project categories, and we have not made an adjustment to the preliminary score based on additional contribution to sustainability considerations.



Preliminary contribution to sustainability

The preliminary contribution to sustainability is significant, based on the relevance and magnitude of the eligible project categories. The issuer shared information about past capital spending allocated to projects that meet at least the substantial contribution criteria of the EU Taxonomy. We have weighted the eligible categories accordingly. A detailed assessment by eligible category is provided below.

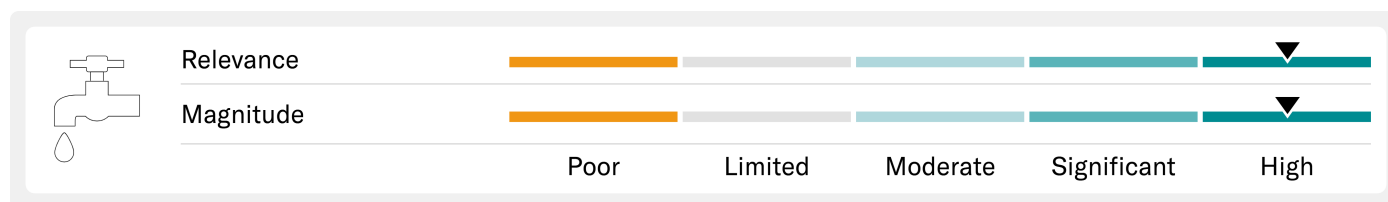
Energy



The overall relevance of the various energy projects financed in this category is significant to address the need for low-carbon energy production development. On one hand, the development of low-carbon heat and electricity, and related infrastructure is highly relevant. According to the International Energy Agency's (IEA) net-zero scenario², two-thirds of global energy in 2050 should come from renewable energy, regardless of the projects' locations. On the other hand, biofuels decarbonization potential can vary across industries; therefore, their relevance in advancing climate mitigation strategies largely depends on the sectors in which they are applied. In this case, it will be mainly used for electricity and heating production, which are relevant uses, although less relevant than the production of biofuels for transport. In addition, financing biomass projects in countries potentially exposed to emissions linked to land use change can be detrimental to climate change mitigation objectives and are therefore considered less relevant. Bioenergy projects implemented in countries with historical emissions from agriculture, forestry, and other land use (AFOLU), which accounted for 13%-20% of total GHG emissions in the last decade³, may have a negative impact on the stated objective of climate change mitigation.

The projects have significant magnitude because of their potential to reduce GHG emissions linked with heat and electricity production. The production of electricity from solar panels, whether mounted on roofs or on the ground, with potential risks assessed through an environmental impact assessment, is considered to be aligned with the best available standards. This is also the case for electricity transmission and distribution assets, which will be externally audited against the EU Taxonomy's substantial contribution criteria. The biomass to produce electricity and heat aligns with the EU Taxonomy's substantial contribution criteria, which also requires a substantial GHG emissions reduction of at least 80%. However, it also allows for woody biomass sourcing, which can entail potential negative externalities. District heating aligned with directive 2012/27/EU is considered to provide benefits compared with standard heating systems. However, because this directive allows up to 50% of heat to be produced from coal, it is not recognized as the best available standard.

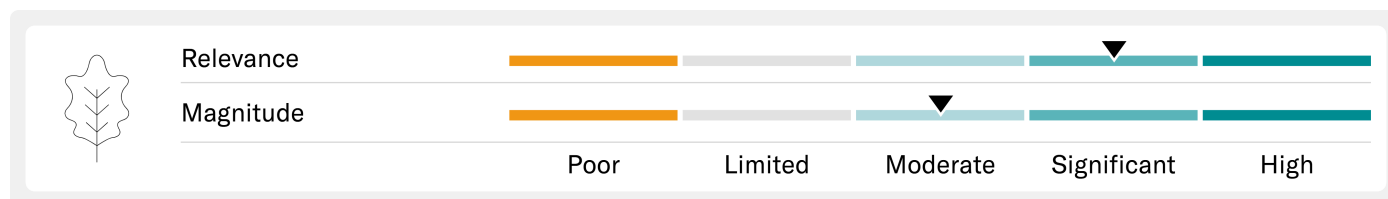
Water



The construction and operation of water treatment and supply systems, and wastewater management facilities are highly relevant to address the increasing water scarcity faced by urban populations globally. This is particularly important because more than 733 million people live in countries experiencing high or critical water stress⁴. The share of global urban population facing water scarcity is projected to rise significantly by 2050. These projects are highly relevant to Veolia, a key company in the water and wastewater management sector, with a significant portion of its business revenue coming from water-related activities. By ensuring access to clean drinking water and effective sanitation services, Veolia supports human health, agriculture and industrial activities, ultimately contributing to overall GDP growth.

The projects have high magnitude because of their potential positive environmental impacts. Veolia aims to comply with the EU Taxonomy's substantial contribution criteria for the economic activities contribution to climate change mitigation (CCM) 5.1 and CCM 5.3. The projects will aim to enhance urban wastewater infrastructure and water supply systems by limiting the energy consumption per cubic meter or per population equivalent or limiting leakages of the assets financed. These are considered as following the best available standards in the market. The company has excluded desalination plants from financing, that usually entail negative environmental externalities.

Waste



The expansion of sustainable waste management practices, which is a key activity for Veolia, is significantly relevant to addressing the challenges associated with increasing generation of waste and its environmental impact. This is particularly important because global waste generation is projected to rise to 3.4 billion tonnes by 2050 from 2.01 billion tonnes in 2016⁵. However, the relevance of these projects varies widely based on the geographic context in which sustainable waste management practices are adopted. In low-income countries, where waste collection and recycling rates are low, these projects might have higher relevance than in high-income regions with advanced waste management systems.

The magnitude of the waste projects financed is overall moderate due to a mix of projects with clear environmental benefits and others with considerable negative environmental externalities. For the collection and transport of waste, Veolia strives to follow the EU Taxonomy's criteria CCM 5.5, transition to a circular economy (CE) 2.3, and pollution prevention and control (PPC) 2.1. These projects will primarily finance trucks and waste boxes, with mixed environmental impacts depending on the type of technology selected. Although the financed trucks will be either electric, liquefied natural gas (LNG), or gas-to-liquids (GTL) trucks, combustion engines are not the best available technology and entail continuous GHG emissions.

For non-hazardous waste management, Veolia aims to follow the EU Taxonomy's criteria CCM 5.09 and CE 2.07, which solely entail mechanical recycling considered the best available standard. For composting of bio-waste, the issuer ensures that methane leakages are monitored and controlled, and that the compost is used as a fertilizer or soil improver, aligning with the best available standard. Gas capture from closed landfills helps reduce GHG emissions, with projects respecting a minimum 70% capture rate, which is considered good, although not the best threshold available in the market. With current criteria, landfill projects are considered to have a moderate magnitude on their own.

For hazardous waste management, Veolia aims to follow the EU Taxonomy's criteria CE 2.4 and PPC 2.2, which involve the treatment and recovery of various hazardous wastes. These activities handle potentially toxic products and use methods that can potentially harm the environment. However, they are expected to yield benefits, with minimum thresholds established by Veolia to ensure environmental safety and sustainability.

Additional contribution to sustainability considerations

We have not made an adjustment to the preliminary contribution to sustainability score based on additional considerations.

We consider that the company has allocated appropriate resources to identify and manage E&S risks. Veolia requires each project to comply with its environmental and social requirements, emphasizing its commitment to environmental stewardship through initiatives like GreenUp, which focuses on depollution and decarbonization targets, and Veolia Cares, which ensures social protection for employees with benefits such as parental leave and health coverage. However, there is an issue with Veolia's operation of the Yerbabuena landfill in Colombia in 2023, which might have led to environmental and health impacts on the local community, including water and food source contamination. The company has drawn criticism for its failure to address these issues and for neglecting the risks faced by environmental defenders. Despite this case, we have decided to not notch down the framework for ESG risk considerations, because it seems like an isolated case. The company has confirmed that it will abide by national regulatory standards, and in their absence, will conduct internal risk assessments. Veolia has also committed to ensure no degradation of high-carbon stock lands, preventing significant effects on Natura 2000 sites, and preventing or managing the spread of invasive alien species.

Regarding Veolia's global sustainability strategy, the categories financed under Veolia's framework are deemed coherent with the company's commitment to sustainability and environmental stewardship, focusing on water management, waste management and energy services. As a global leader in optimized resource management, Veolia helps communities and industries to reduce their environmental footprint and enhance resource efficiency, supporting global efforts towards circular economy.

Appendix 1 - Alignment with principles scorecard for Veolia's green financing framework

Factor	Sub-factor	Component	Component score	Sub-factor score	Factor score
Use of proceeds	Clarity of the eligible categories	Nature of expenditure	A	Best practices	Best practices
		Definition of content, eligibility and exclusion criteria for nearly all categories	A		
		Location	A		
		BP: Definition of content, eligibility and exclusion criteria for all categories	Yes		
	Clarity of the objectives	Relevance of objectives to project categories for nearly all categories	A	Best practices	
		Coherence of project category objectives with standards for nearly all categories	A		
		BP: Objectives are defined, relevant and coherent for all categories	Yes		
	Clarity of expected benefits	Identification and relevance of expected benefits for nearly all categories	A	Best practices	
		Measurability of expected benefits for nearly all categories	A		
		BP: Relevant benefits are identified for all categories	Yes		
		BP: Benefits are measurable for all categories	Yes		
		BP: Disclosure of refinancing prior to issuance and in post-allocation reporting	Yes		
		BP: Commitment to communicate refinancing look-back period prior to issuance	Yes		
Process for project evaluation and selection	Transparency and clarity of the process for defining and monitoring eligible projects	Clarity of the process	A	Best practices	Best practices
		Disclosure of the process	A		
		Transparency of the environmental and social risk mitigation process	A		
		BP: Monitoring of continued project compliance	Yes		
Management of proceeds	Allocation and tracking of proceeds	Tracking of proceeds	A	Best practices	Best practices
		Periodic adjustment of proceeds to match allocations	A		
		Disclosure of the intended types of temporary placements of unallocated proceeds	A		
		BP: Disclosure of the proceeds management process	Yes		
		BP: Allocation period is 24 months or less	Yes		
Reporting	Reporting transparency	Reporting frequency	A	Aligned	Aligned
		Reporting duration	A		
		Report disclosure	A		
		Reporting exhaustivity	A		
		BP: Allocation reporting at least until full allocation of proceeds, and impact reporting until full bond maturity or loan payback	No		
		BP: Clarity and relevance of the indicators on the sustainability benefits	Yes		
		BP: Disclosure of reporting methodology and calculation assumptions	Yes		
		BP: Independent external auditor, or other third party, to verify the tracking and allocation of funds	Yes		
		BP: Independent impact assessment on environmental and social benefits	No		
Overall alignment with principles score:				Aligned	

Appendix 2 - Mapping eligible categories to the United Nations' Sustainable Development Goals

The three eligible categories included in Veolia's green financing framework are likely to contribute to five of the UN's SDGs, namely:

UN SDG 17 Goals		SDG Targets
GOAL 3: Good Health and Well-being	Waste	3.9: Substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution
		6.1: Achieve universal and equitable access to safe and affordable drinking water for all
GOAL 6: Clean Water and Sanitation	Water	6.2: Achieve access to adequate sanitation and hygiene for all and end open defecation
		6.3: Improve water quality by reducing pollution, eliminating dumping and minimizing hazardous chemicals and materials
GOAL 7: Affordable and Clean Energy	Energy	7.2: Increase substantially the share of renewable energy in the global energy mix
GOAL 11: Sustainable Cities and Communities	Waste	11.1: Ensure access for all to adequate, safe and affordable housing and basic services and upgrade slums
GOAL 12: Responsible Consumption and Production	Waste	12.4: Achieve environmental management of chemicals and all wastes, and reduce their release to air, water and soil
GOAL 13: Climate Action	Energy, Water and Waste	13.1: Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters in all countries

The mapping of the UN's SDGs in this SPO considers the eligible project categories and associated sustainability objectives/benefits documented in the issuer's green financing framework, as well as resources and guidelines from public institutions, such as the ICMA's SDG Mapping Guidance and the UN's SDG targets and indicators.

Appendix 3 - Summary of eligible categories in Veolia's framework

Eligible Category	Sub-category	Description	Sustainability Objective	Impact Reporting Metrics
Energy	Production of heat/cool and district heating/cooling distribution	District heating/cooling: Construction, refurbishment and operation of pipelines and associated infrastructure for distribution of heating and cooling Construction and operation of facilities that produce heat/cool from bioenergy or waste heat EU Taxonomy associated categories: CCM 4.15, 4.20, 4.24, 4.25.	Climate change mitigation	- Energy generated (thermal and electrical) - Share of renewable and recovered - Greenhouse gas emission savings from the use of biomass
	Electricity generation from bioenergy	Construction and operation of electricity generation installations that produce electricity exclusively from Biomass (including the biogenic fraction of municipal waste to be incinerated), biogas or bioliquids, excluding electricity generation from blending of renewable fuels with biogas or bioliquids. EU Taxonomy associated category: CCM 4.8		- Greenhouse gas emission savings from the use of biomass - Energy generated (thermal and electrical)
	Transmission and distribution of electricity	Construction and operation of transmission and distribution systems that transport electricity EU Taxonomy associated category: CCM 4.9		- Newly enabled generation capacity in the system - Average system grid emissions factor - GHG emissions avoided/reduced
	Solar Photovoltaic electricity production	Installation and maintenance of solar photovoltaic panels This activity is to be used for installations dedicated to PV electricity generation, such as PV farms on landfills or WWTP EU Taxonomy associated category: CCM 4.1		- Solar PV installed capacity - GHG emission avoided
Water	Freshwater supply	Activities of water abstraction and/or treatment plant and/or drinking water distribution and/or production of industrial water EU Taxonomy associated categories: CCM 5.1	Climate Change Mitigation Sustainable use and protection of water and marine resources	- Amount of drinking water produced and distributed - Energy efficiency of drinking water production (excluding seawater desalination) (pro forma 2019-2023) - Net average energy consumption for abstraction and treatment
	Wastewater treatment	Activities of sewer networks and/or wastewater treatment plants EU Taxonomy associated categories: CCM 5.3	Sustainable use and protection of water and marine resources	- Energy efficiency of wastewater treatment: Net energy consumption of the wastewater treatment plant - Volume of water reused from collected and treated wastewater - Wastewater treatment sludge recovery rate

Eligible Category	Sub-category	Description	Sustainability Objective	Impact Reporting Metrics
Waste	Collection and transport of waste	Waste collection services for all types of client (municipal, commercial, or industrial), for non-hazardous waste in single or comingled fractions aimed at preparing for reuse or recycling. Where financing is directed towards collection vehicles, these will be limited to those that are either fully electric or powered by LNG or gas-to-liquids (GTL). Separate collection and transport of hazardous waste prior to treatment, material recovery or disposal. EU Taxonomy associated categories: CCM 5.5, CE 2.3, PPC 2.1	Climate Change Mitigation Circular Economy Pollution Prevention and Control	- Municipal and industrial waste collected (t) - Amount of hazardous waste collected/treated - Number of financed collection vehicles (and types of vehicles) - Type of financed waste storage
	Non hazardous waste management	Construction and operation of facilities for the sorting and processing of separately collected non-hazardous waste streams into secondary raw materials involving mechanical reprocessing, except for backfilling purpose. Construction and operation of dedicated facilities for the treatment of separately collected bio-waste through composting (aerobic digestion) with the resulting production and utilisation of compost. Installation and operation of infrastructure for landfill gas capture and utilisation in permanently closed landfills or landfill cells using new or supplementary dedicated technical facilities and equipment installed during or post landfill or landfill cell closure. Minimum gas recovery of 70% EU Taxonomy associated categories: CCM 5.09, 5.8, 5.10, CE 2.7		- Volume of treated waste - Volume of plastics recycled in Veolia's processing plants - Material recovery rate from treated waste - Amount of waste composted
	Hazardous waste management	Construction, repurposing, upgrade, and operation of dedicated facilities for the treatment of hazardous waste, including as a means for material recovery operations EU Taxonomy associated categories: CE 2.4, PPC 2.2		- Material recovery rate of secondary raw materials (including chemical substances and critical raw materials) meant to substitute primary raw materials or chemicals in production processes

Endnotes

- [1](#) The point-in-time assessment is applicable only on the date of assignment or update.
- [2](#) [International Energy Agency Net Zero Scenario](#).
- [3](#) [Intergovernmental Panel on Climate Change 6th assessment report](#)
- [4](#) [United Nations Statistics Division 2022 report](#).
- [5](#) [What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050](#), World Bank report.

Moody's assigns SPOs in alignment with the main tenets of the ICMA Guidelines for Green, Social, Sustainability and Sustainability-Linked Bonds External Reviews and the LSTA/LMA/APLMA Guidance for Green, Social and Sustainability-Linked Loans External Reviews, as applicable; Moody's practices may however diverge in some respects from the practices recommended in those documents. Moody's approach to assigning SPOs is described in its Assessment Framework, and is subject to the ethical and professional principles set forth in the Moody's Investors Service Code of Professional Conduct.

Additional terms with respect to Second Party Opinions (as defined in Moody's Investors Service Rating Symbols and Definitions): Please note that a Second Party Opinion ("SPO") is not a "credit rating". The issuance of SPOs is not a regulated activity in many jurisdictions, including Singapore. JAPAN: In Japan, development and provision of SPOs fall under the category of "Ancillary Businesses", not "Credit Rating Business", and are not subject to the regulations applicable to "Credit Rating Business" under the Financial Instruments and Exchange Act of Japan and its relevant regulation. PRC: Any SPO: (1) does not constitute a PRC Green Bond Assessment as defined under any relevant PRC laws or regulations; (2) cannot be included in any registration statement, offering circular, prospectus or any other documents submitted to the PRC regulatory authorities or otherwise used to satisfy any PRC regulatory disclosure requirement; and (3) cannot be used within the PRC for any regulatory purpose or for any other purpose which is not permitted under relevant PRC laws or regulations. For the purposes of this disclaimer, "PRC" refers to the mainland of the People's Republic of China, excluding Hong Kong, Macau and Taiwan.

© 2025 Moody's Corporation, Moody's Investors Service, Inc., Moody's Analytics, Inc. and/or their licensors and affiliates (collectively, "MOODY'S"). All rights reserved.

CREDIT RATINGS ISSUED BY MOODY'S CREDIT RATINGS AFFILIATES ARE THEIR CURRENT OPINIONS OF THE RELATIVE FUTURE CREDIT RISK OF ENTITIES, CREDIT COMMITMENTS, OR DEBT OR DEBT-LIKE SECURITIES, AND MATERIALS, PRODUCTS, SERVICES AND INFORMATION PUBLISHED OR OTHERWISE MADE AVAILABLE BY MOODY'S (COLLECTIVELY, "MATERIALS") MAY INCLUDE SUCH CURRENT OPINIONS. MOODY'S DEFINES CREDIT RISK AS THE RISK THAT AN ENTITY MAY NOT MEET ITS CONTRACTUAL FINANCIAL OBLIGATIONS AS THEY COME DUE AND ANY ESTIMATED FINANCIAL LOSS IN THE EVENT OF DEFAULT OR IMPAIRMENT. SEE APPLICABLE MOODY'S RATING SYMBOLS AND DEFINITIONS PUBLICATION FOR INFORMATION ON THE TYPES OF CONTRACTUAL FINANCIAL OBLIGATIONS ADDRESSED BY MOODY'S CREDIT RATINGS. CREDIT RATINGS DO NOT ADDRESS ANY OTHER RISK, INCLUDING BUT NOT LIMITED TO: LIQUIDITY RISK, MARKET VALUE RISK, OR PRICE VOLATILITY. CREDIT RATINGS, NON-CREDIT ASSESSMENTS ("ASSESSMENTS"), AND OTHER OPINIONS INCLUDED IN MOODY'S MATERIALS ARE NOT STATEMENTS OF CURRENT OR HISTORICAL FACT. MOODY'S MATERIALS MAY ALSO INCLUDE QUANTITATIVE MODEL-BASED ESTIMATES OF CREDIT RISK AND RELATED OPINIONS OR COMMENTARY PUBLISHED BY MOODY'S ANALYTICS, INC. AND/OR ITS AFFILIATES. MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS DO NOT CONSTITUTE OR PROVIDE INVESTMENT OR FINANCIAL ADVICE, AND MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS ARE NOT AND DO NOT PROVIDE RECOMMENDATIONS TO PURCHASE, SELL, OR HOLD PARTICULAR SECURITIES. MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS DO NOT COMMENT ON THE SUITABILITY OF AN INVESTMENT FOR ANY PARTICULAR INVESTOR. MOODY'S ISSUES ITS CREDIT RATINGS, ASSESSMENTS AND OTHER OPINIONS AND PUBLISHES OR OTHERWISE MAKES AVAILABLE ITS MATERIALS WITH THE EXPECTATION AND UNDERSTANDING THAT EACH INVESTOR WILL, WITH DUE CARE, MAKE ITS OWN STUDY AND EVALUATION OF EACH SECURITY THAT IS UNDER CONSIDERATION FOR PURCHASE, HOLDING, OR SALE.

MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS, AND MATERIALS ARE NOT INTENDED FOR USE BY RETAIL INVESTORS AND IT WOULD BE RECKLESS AND INAPPROPRIATE FOR RETAIL INVESTORS TO USE MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS OR MATERIALS WHEN MAKING AN INVESTMENT DECISION. IF IN DOUBT YOU SHOULD CONTACT YOUR FINANCIAL OR OTHER PROFESSIONAL ADVISER.

ALL INFORMATION CONTAINED HEREIN IS PROTECTED BY LAW, INCLUDING BUT NOT LIMITED TO, COPYRIGHT LAW, AND NONE OF SUCH INFORMATION MAY BE COPIED OR OTHERWISE REPRODUCED, REPACKAGED, FURTHER TRANSMITTED, TRANSFERRED, DISSEMINATED, REDISTRIBUTED OR RESOLD, OR STORED FOR SUBSEQUENT USE FOR ANY SUCH PURPOSE, IN WHOLE OR IN PART, IN ANY FORM OR MANNER OR BY ANY MEANS WHATSOEVER, BY ANY PERSON WITHOUT MOODY'S PRIOR WRITTEN CONSENT. FOR CLARITY, NO INFORMATION CONTAINED HEREIN MAY BE USED TO DEVELOP, IMPROVE, TRAIN OR RETRAIN ANY SOFTWARE PROGRAM OR DATABASE, INCLUDING, BUT NOT LIMITED TO, FOR ANY ARTIFICIAL INTELLIGENCE, MACHINE LEARNING OR NATURAL LANGUAGE PROCESSING SOFTWARE, ALGORITHM, METHODOLOGY AND/OR MODEL.

MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS ARE NOT INTENDED FOR USE BY ANY PERSON AS A BENCHMARK AS THAT TERM IS DEFINED FOR REGULATORY PURPOSES AND MUST NOT BE USED IN ANY WAY THAT COULD RESULT IN THEM BEING CONSIDERED A BENCHMARK.

All information contained herein is obtained by MOODY'S from sources believed by it to be accurate and reliable. Because of the possibility of human or mechanical error as well as other factors, however, all information contained herein is provided "AS IS" without warranty of any kind. MOODY'S adopts all necessary measures so that the information it uses in assigning a credit rating is of sufficient quality and from sources MOODY'S considers to be reliable including, when appropriate, independent third-party sources. However, MOODY'S is not an auditor and cannot in every instance independently verify or validate information received in the credit rating process or in preparing its Materials.

To the extent permitted by law, MOODY'S and its directors, officers, employees, agents, representatives, licensors and suppliers disclaim liability to any person or entity for any indirect, special, consequential, or incidental losses or damages whatsoever arising from or in connection with the information contained herein or the use of or inability to use any such information, even if MOODY'S or any of its directors, officers, employees, agents, representatives, licensors or suppliers is advised in advance of the possibility of such losses or damages, including but not limited to: (a) any loss of present or prospective profits or (b) any loss or damage arising where the relevant financial instrument is not the subject of a particular credit rating assigned by MOODY'S.

To the extent permitted by law, MOODY'S and its directors, officers, employees, agents, representatives, licensors and suppliers disclaim liability for any direct or compensatory losses or damages caused to any person or entity, including but not limited to by any negligence (but excluding fraud, willful misconduct or any other type of liability that, for the avoidance of doubt, by law cannot be excluded) on the part of, or any contingency within or beyond the control of, MOODY'S or any of its directors, officers, employees, agents, representatives, licensors or suppliers, arising from or in connection with the information contained herein or the use of or inability to use any such information.

NO WARRANTY, EXPRESS OR IMPLIED, AS TO THE ACCURACY, TIMELINESS, COMPLETENESS, MERCHANTABILITY OR FITNESS FOR ANY PARTICULAR PURPOSE OF ANY CREDIT RATING, ASSESSMENT, OTHER OPINION OR INFORMATION IS GIVEN OR MADE BY MOODY'S IN ANY FORM OR MANNER WHATSOEVER.

Moody's Investors Service, Inc., a wholly-owned credit rating agency subsidiary of Moody's Corporation ("MCO"), hereby discloses that most issuers of debt securities (including corporate and municipal bonds, debentures, notes and commercial paper) and preferred stock rated by Moody's Investors Service, Inc. have, prior to assignment of any credit rating, agreed to pay Moody's Investors Service, Inc. for credit ratings opinions and services rendered by it. MCO and all MCO entities that issue ratings under the "Moody's Ratings" brand name ("Moody's Ratings"), also maintain policies and procedures to address the independence of Moody's Ratings' credit ratings and credit rating processes. Information regarding certain affiliations that may exist between directors of MCO and rated entities, and between entities who hold credit ratings from Moody's Investors Service, Inc. and have also publicly reported to the SEC an ownership interest in MCO of more than 5%, is posted annually at ir.moody.com under the heading "Investor Relations — Corporate Governance — Charter and Governance Documents - Director and Shareholder Affiliation Policy."

Moody's SF Japan K.K., Moody's Local AR Agente de Calificación de Riesgo S.A., Moody's Local BR Agência de Classificação de Risco LTDA, Moody's Local MX S.A. de C.V., I.C.V., Moody's Local PE Clasificadora de Riesgo S.A., and Moody's Local PA Clasificadora de Riesgo S.A. (collectively, the "Moody's Non-NRSRO CRAs") are all indirectly wholly-owned credit rating agency subsidiaries of MCO. None of the Moody's Non-NRSRO CRAs is a Nationally Recognized Statistical Rating Organization.

Additional terms for Australia only: Any publication into Australia of this document is pursuant to the Australian Financial Services License of MOODY'S affiliate, Moody's Investors Service Pty Limited ABN 61 003 399 657 AFSL 336969 and/or Moody's Analytics Australia Pty Ltd ABN 94 105 136 972 AFSL 383569 (as applicable). This document is intended to be provided only to "wholesale clients" within the meaning of section 761G of the Corporations Act 2001. By continuing to access this document from within Australia, you represent to MOODY'S that you are, or are accessing the document as a representative of, a "wholesale client" and that neither you nor the entity you represent will directly or indirectly disseminate this document or its contents to "retail clients" within the meaning of section 761G of the Corporations Act 2001. MOODY'S credit rating is an opinion as to the creditworthiness of a debt obligation of the issuer, not on the equity securities of the issuer or any form of security that is available to retail investors.

Additional terms for India only: Moody's credit ratings, Assessments, other opinions and Materials are not intended to be and shall not be relied upon or used by any users located in India in relation to securities listed or proposed to be listed on Indian stock exchanges.

Additional terms with respect to Second Party Opinions and Net Zero Assessments (as defined in Moody's Ratings Rating Symbols and Definitions): Please note that neither a Second Party Opinion ("SPO") nor a Net Zero Assessment ("NZA") is a "credit rating". The issuance of SPOs and NZAs is not a regulated activity in many jurisdictions, including Singapore.

JAPAN: In Japan, development and provision of SPOs and NZAs fall under the category of "Ancillary Businesses", not "Credit Rating Business", and are not subject to the regulations applicable to "Credit Rating Business" under the Financial Instruments and Exchange Act of Japan and its relevant regulation. PRC: Any SPO: (1) does not constitute a PRC Green Bond Assessment as defined under any relevant PRC laws or regulations; (2) cannot be included in any registration statement, offering circular, prospectus or any other documents submitted to the PRC regulatory authorities or otherwise used to satisfy any PRC regulatory disclosure requirement; and (3) cannot be used within the PRC for any regulatory purpose or for any other purpose which is not permitted under relevant PRC laws or regulations. For the purposes of this disclaimer, "PRC" refers to the mainland of the People's Republic of China, excluding Hong Kong, Macau and Taiwan.

REPORT NUMBER 1440212